

MANISH RAINA

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PROFILE

AI/ML Engineer with hands-on, end-to-end project experience across model development, backend engineering, and full-stack delivery. Comfortable building with Python, Scikit-learn, TensorFlow, FastAPI/React, Docker, and RAG/MCP-based agent systems. Looking to bring strong fundamentals and a builder's mindset to a production ML or AI engineering team.

EDUCATION

B.Tech, Computer Science Engineering (AI & ML) — PIET, Haryana

Aug 2022 – Aug 2026

Senior Secondary (Class XII) — Kotwal National Institute

Mar 2018 – Mar 2022

CORE COMPETENCIES

- **Languages:** Python, C, C++, JavaScript, SQL, TypeScript
- **AI Agents & Orchestration:** LangChain, LangGraph, Model Context Protocol (MCP), Multi-Agent Systems, Agentic Tool/Skill Design, Function Calling, Prompt Engineering, RAG Pipelines
- **ML & Model Training:** Scikit-learn, TensorFlow, LightGBM, NLP, Model Training & Fine-Tuning, Embedding-based Semantic Search, Model Evaluation (AUC, Cross-Validation, Calibration, IPCW Weighting)
- **Web & Backend / APIs:** FastAPI, React.js, REST APIs, API Design & Integration, WebSockets, JWT Authentication, RBAC, Middleware Validation
- **Data & Infra:** PostgreSQL, MongoDB, Redis, SQLite, Qdrant Vector DB, Docker, Docker-compose, Git
- **Other:** Web Scraping, Power BI, Speech Recognition & Synthesis (SpeechRecognition, pyttsx3), OpenCV, dlib, Playwright

EXPERIENCE

Machine Learning & Web Development Intern — EISystems Technologies

Aug 2024 – Sept 2024

- Worked on data preprocessing and ML model development for client projects, alongside front-end feature development using React.js.

Machine Learning Research Intern — NIT Delhi

Aug 2025 – Sept 2025

- Researched and implemented supervised machine learning algorithms using Python, Scikit-learn, Pandas, and NumPy; conducted comparative performance analysis and prepared technical documentation of experimental results.

PROJECTS

Nexus AI OS v2.0 — Autonomous Multi-Agent Workspace & AI Orchestration Platform

[\[Github\]](#)

- Engineered a custom multi-agent orchestration kernel that routes tasks through 12 specialized agents via a JSON-based DAG, with planner, reflection, and supervisor safety layers.
- Built a custom MCP client with stdio JSON-RPC tool discovery, a RAG pipeline indexed via Qdrant, and a hybrid PostgreSQL/MongoDB/Redis storage layer with human-in-the-loop safety gates.

AI Career Skill Intelligence Platform (Full-Stack + ML + Scraper)

[\[Github\]](#)

- Designed and built an end-to-end career intelligence platform using FastAPI, React, and PostgreSQL, implementing NLP-based resume parsing and skill-extraction pipelines for candidate profiles.
- Implemented embedding-based semantic matching between job listings and candidate profiles, and built a web scraper with a dashboard visualizing in-demand skill trends.

Heart Disease Survival Prediction Framework

[\[Github\]](#)

- Built a survival prediction model using LightGBM with IPCW weighting to handle censored clinical outcome data from a heart disease dataset.
- Applied Scikit-learn for cross-validation and calibration, using AUC-based evaluation to rigorously assess model discrimination and generalization across held-out test folds.

AI Voice Receptionist (Speech-to-Speech AI Assistant)

[\[Github\]](#)

- Built a voice assistant pipeline using SpeechRecognition and PyAudio for input capture, with pyttsx3 for natural speech output and response generation.
- Integrated OpenCV, dlib, and face_recognition for a real-time face-recognition layer, alongside rule-based conversation logic for automated query handling.

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning with Python — Internship Certificate, EISystems Technologies
- Machine Learning Research Internship — NIT Delhi (2025)
- Advanced Diploma in Software Technology (12 Months) — Supertech (India) Computer Education